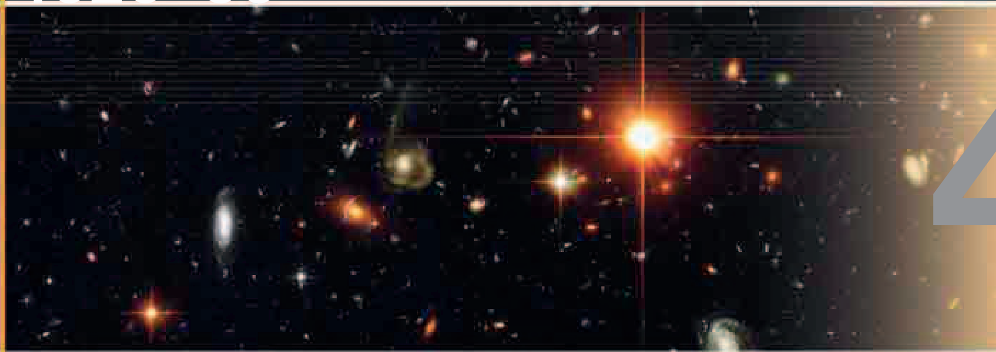


46 BILLION THE SIZE OF THE UNIVERSE VISIBLE FROM EARTH IN LIGHT-YEARS

This image taken by the Hubble Space Telescope reveals some of the most distant galaxies in the Universe. They are billions of light-years beyond the foreground stars.

BY 1920 ASTRONOMERS HAD REALIZED that the Universe was not centered on our Milky Way, but rather that the **Milky Way was a single galaxy** among many others. As the decade passed, another still more extraordinary revelation came to light. It appeared that the **Universe** did not have a fixed finite size. Astronomers had discovered that the light spectrum of stars carried wavelengths that were "stretching out" toward the red end of the spectrum—which they **called redshift** (see panel, right). This redshift indicated that these stars were moving away. In other words, **the Universe is expanding**. In 1929, US astronomer **Edwin Hubble** (1889–1953) explained this relationship and noted that the most **distant galaxies were receding at the fastest rate**. This later became known as **Hubble's Law**.

Television technology took another step forward in June 1929 when the **Bell Laboratory** in the US demonstrated the **first color image** transmission. The subjects they used were chosen for maximum impact: a woman with flowers and the US flag.

In April 1930, US chemist **Elmer Bolton** (1886–1968) made one of the first synthetic rubbers. A derivative of acetylene, it was later called **neoprene**. More corrosion-resistant than natural rubber, neoprene was suitable for use in extreme conditions, such as for hoses and fire-resistant coatings.

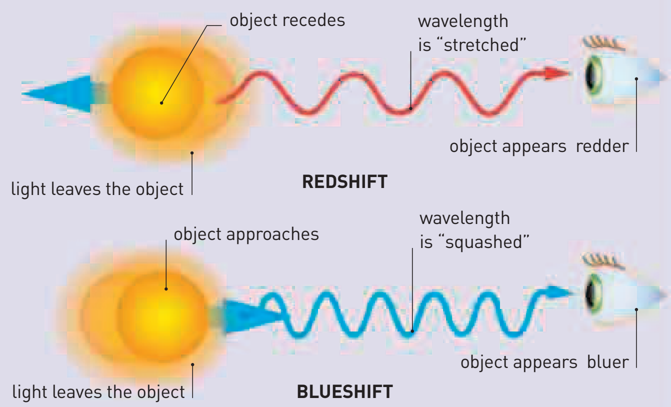
Photographs taken at the Lowell Observatory in Arizona in March 1917 had recorded the faint image of a body that became known as "X." In February 1930, American astronomer **Clyde Tombaugh** (1906–97) confirmed that **X was a planet**, and in May, it was given its name: **Pluto**.

Two years after Alexander Fleming made his accidental discovery of penicillin, this new antibiotic had its first curative use. British physician **Cecil George Paine** (1905–94), a former student of Fleming, had taken samples of the mold to his workplace in Sheffield, UK. In August 1930, he successfully used its filtrate to **treat eye infections**.

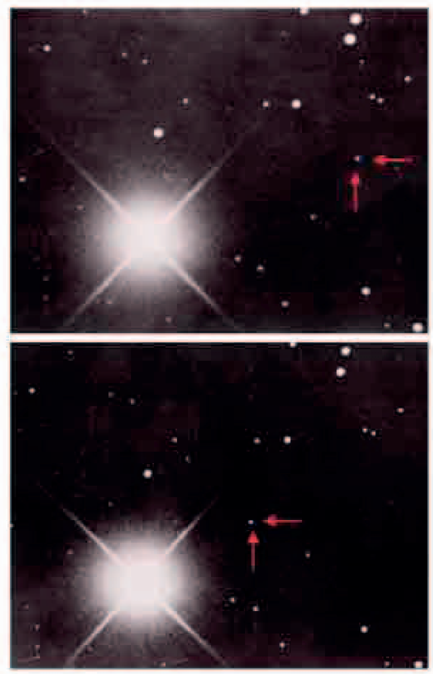
Paul Dirac (see panel, opposite) used quantum physics theory to correctly predict the existence of antimatter. In 1930, he published ***Principles of Quantum Mechanics***, which for decades would be the standard textbook.

REDSHIFT AND BLUESHIFT

The movement of a light-emitting object, such as a galaxy, affects its visible light spectrum. If an object is moving away, its wavelengths appear stretched out. As longer wavelengths belong to red light, this is described as a redshift. Movement toward the observer squashes up the wavelengths, giving a blueshift. Galaxies exhibit a redshift, so they are moving away.



First color television demonstration
Open doors reveal the working parts. Light is directed at the operator through a scanning disk (center) onto photoelectric cells (left).



Existence of Pluto
These two photographs of the sky taken in 1930 on different nights show a "body" (arrowed) that has changed position, which indicates that it is closer than surrounding stars. The body was a planet and was given the name Pluto.

January 1929 Czechoslovakian-born biochemists **Carl and Gerty Cori** describe how **glucose** is changed to **lactic acid** during exercise

March 15, 1929 Edwin Hubble describes evidence that the **Universe is expanding** (Hubble's Law)

June 27, 1929 Bell publically demonstrates **color television**

September 1, 1929 British and Dutch-born physicists **Robert Atkinson and Fritz Houtermans** explain how **energy from stars** comes from **nuclear fusion**

December 1929 America astronomer **Andrew Douglass** describes a **tree-ring dating** method used at archaeological sites

January 1, 1930 Paul Dirac describes **electron holes**, later called **positrons**

February 18, 1930 Clyde Tombaugh confirms the existence of **Planet X**

April 17, 1930 Elmer Bolton makes a **synthetic rubber**

November 25, 1930 Cecil George Paine reports the **first cure** using **penicillin**

December 4, 1930 Austrian physicist **Wolfgang Pauli** proposes existence of **neutrinos**